SQL Saturday #196

SQL Server 2012 AlwaysOn

A big thanks to our sponsors

Speaker

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- Consultant and mentor at SolidQ
  - Administration, capacity planning, data modeling, development, optimization
  - >15 years on SQL Server (from 4.21 in 1996)
- Interests
  - Relational model, DBMS architectures, high availability and disaster recovery
- Microsoft MVP SQL Server since 1998
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AlwaysOn

▶ Introduction

- New name for HA and DR technologies
  - Failover Cluster Instance (FCI)
  - Availability Groups (AG)
- Previously named HADRON
  - Keep in mind when searching articles
  - Still present in some DMVs (_hadr_)
- For the purpose of this presentation
  - High Availability (local HA): in a data center
  - Disaster Recovery: across data centers

AlwaysOn FCI

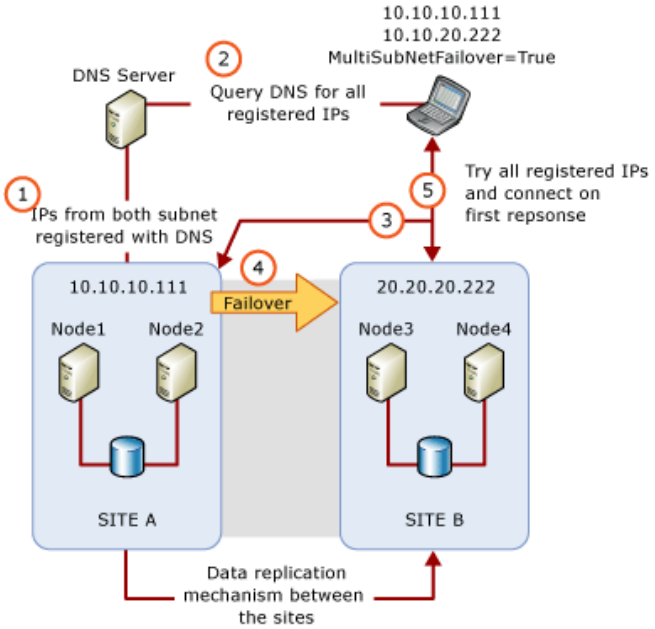
▶ Brief review

- Protects an entire SQL Server instance
- Requires Shared Storage
- Geographic DR with SQL Server < 2012
  - Proprietary solutions for Stretch V-LAN
  - Proprietary solutions for storage replication

AlwaysOn FCI

▶ Multi-Subnet Support

- Nodes on different subnets
  - IP Address resources in OR
- Still need storage replication
- Enterprise Edition required



http://msdn.microsoft.com/en-us/library/ff878716.aspx

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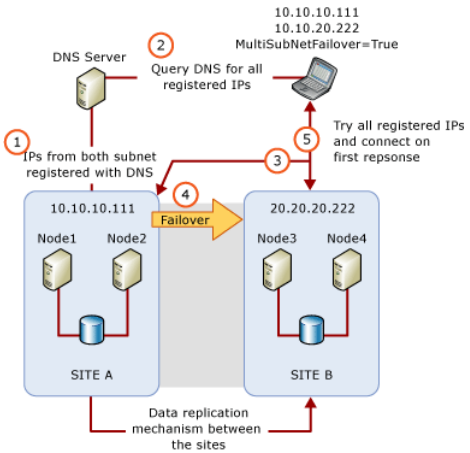
▶ Multi-Subnet Support: how do clients reconnect?

- New clients
  - Try IP addresses in order
  - Avoid DNS update latencies
  - New connection string parameter
    - MultiSubnetFailover=True
    - Triy IP addresses in parallel
- Old clients
  - Change timeout value
  - Reduce DNS updates latency (HostRecordTTL)
  - Reduce DNS cache expiration on the client
- Check driver support!
  - http://msdn.microsoft.com/en-us/library/hh995351(v=sql.15).aspx



http://msdn.microsoft.com/en-us/library/ff878716.aspx

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▶ Failover policies in SQL Server 2008

- Looks Alive
  - Checks service state (Service Control Mgr.)
  - Every 5 seconds
- IsAlive
  - Checks **SELECT @@SERVERNAME**
  - Every 60 seconds
  - Can pass check even when not responding to other incoming requests
    - e.g. stuck schedulers

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▶ Flexible failover policies in SQL Server 2012

- IsAlive based on system procedure
  - Resource DLL calls **sp_server_diagnostic**
    - Over a dedicated connection
  - Diagnostic information from procedure
    - New instance parameter **FailureConditionLevel**
    - Allows selecting conditions for failover to occur
  - Timeouts while calling diagnostic procedure
    - New Cluster parameter **HealthCheckTimeout**

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▶ Failure condition levels

Level	Condition	Description
0	No automatic failover or restart	Never Failover (system maintenance only)
1	Failover or restart on SQL Server down	Service down
2	Failover or restart on SQL Server unresponsive	Server hanged (sp_server_diagnostics not responding within HealthCheckTimeout setting)
3	Failover or restart on critical SQL Server errors	System unhealthy (sp_server_diagnostics “system error”)
4	Failover or restart on moderate SQL Server errors	Resource unhealthy (sp_server_diagnostics “resource error”)
5	Failover or restart on any qualified failure	QP Unhealthy (sp_server_diagnostics “query_processing error”)

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▶ Flexible failover policies configuration

- Failover Cluster Manager
- PowerShell
- T-SQL
  - ALTER SERVER CONFIGURATION FAILOVER CLUSTER PROPERTY HEALTHCHECKTIMEOUT = 60000;
  - ALTER SERVER CONFIGURATION SET FAILOVER CLUSTER PROPERTY FAILURECONDITIONLEVEL = 3;

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▶ Reducing planned downtime with SQL Server 2012

- Windows Core Support
  - Less patches to be installed (circa 50-60%)
- Support for rolling upgrading and patching
- Better control on failover time
  - Indirect Checkpoints
    - ALTER DATABASE …
      SET TARGET_RECOVERY_TIME = …

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▶ New Storage options with SQL Server 2012

- Tempdb on local storage
  - Can leverage fast SSD cards
- SMB shares supported
  - Best on
    - Windows Server 2008 R2 (SMB 2.1 + Hotfix)
    - Windows Server 2012
- Asymmetric Disks

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▶ Data replicas review

- Log Shipping
- Database Mirroring
- Replication
- SQL Server 2012: Availability Groups

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▶ Log Shipping

- Advantages
  - Entire database replicas
  - Multiple replicas per database
  - Simple and robust
- Disadvantages
  - Only asynchronous replicas
  - Readable replicas with trade-offs
    - Updated to last log restore
    - Need to disconnect clients before restoring

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▶ Database Mirroring

- Advantages
  - Entire database replicas
  - Synchronous and asynchronous (EE) replicas
  - Simple and robust
- Disadvantages
  - Only one replica per database
  - Readable replicas need Database Snapshot
    - Enterprise Edition
    - Updated to last snapshot creation time
    - Need to disconnect or handle multiple snapshot names

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▶ Replication

- Advantages
  - More granular (depends on scenario)
  - Multiple replicas
  - Readable & updateable replicas
- Disadvantages
  - More granular (depends on scenario)
  - Only asynchronous replicas
  - Typically higher latency
  - Complexity and maintenance overhead

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▶ Common Problems

- Manual Failover*
- No coordination to failover multiple DBs
- No backups on replicas*
- Specific limitations
  - E.g. FILESTREAM, cross-database consistency of transactions, only one replica

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▶ AG as evolution of Database Mirroring

- Coordinated failover of multiple DBs
- Multiple replicas (up to 4)
- Synchronous/asynchronous replicas
- Up-to-date readable replicas
- Ability to off-load maintenance on replicas e.g. backups

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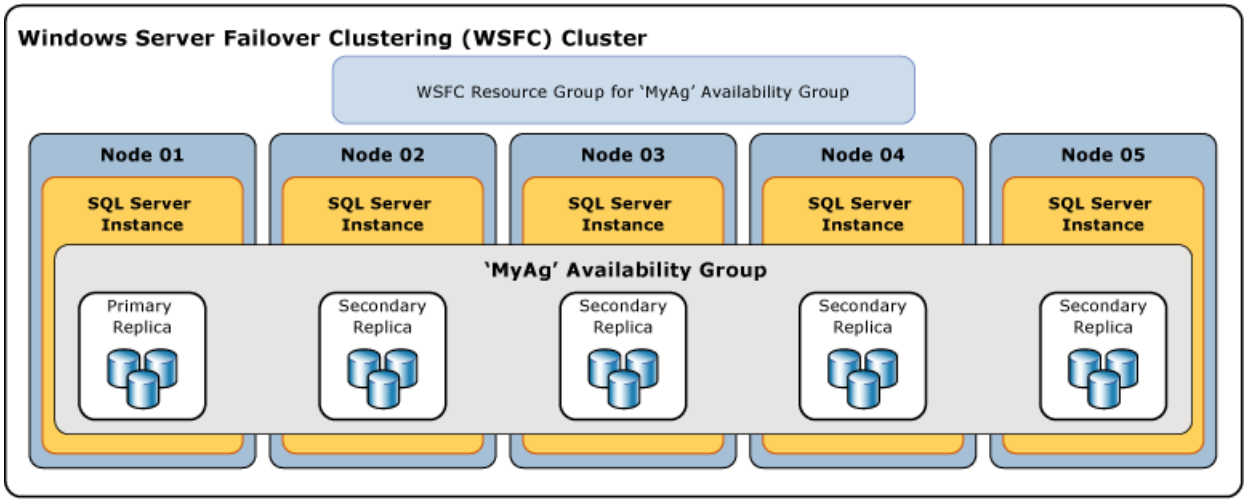
▶ Overview



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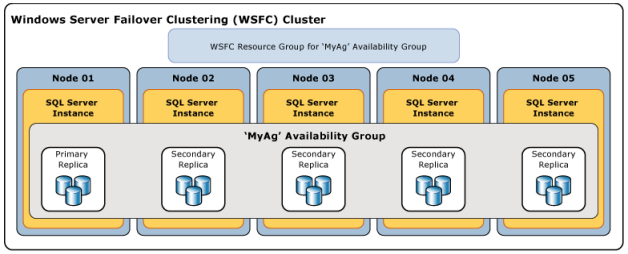
▶ Windows Server Failover Clustering (WSFC)



- Used by Availability Groups for
  - Primary health detection
  - Inter-node health detection
  - Failover coordination
  - Distributed data store for settings and state
  - Distributed change notifications between nodes

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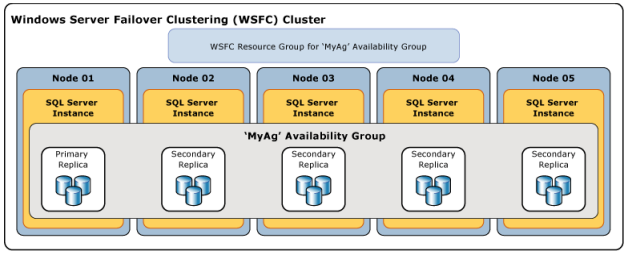
▶ Groups and databases



- Availability Group
  - Set of availability databases failover over together
- Availability Database
  - Database belonging to an availability group
- Primary Database
  - Read/write copy of an availability database
- Secondary Database
  - Read only copy of an availability database

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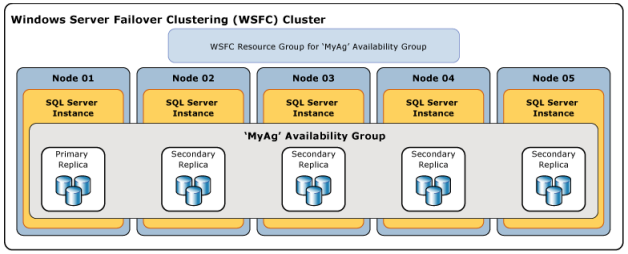
▶ Replicas



- Availability Replica
  - Instance of availability group hosted on a specific SQL Server instance
- Primary Replica
  - Makes primary databases available in read/write
  - Send log records to synchronize secondary replicas
- Secondary Replica
  - Maintains read/only copies of availability DBs

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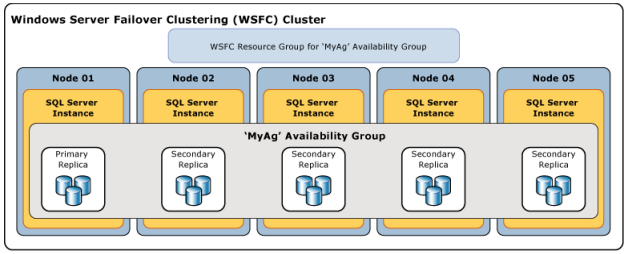
▶ Connectivity



- Availability Group Listener
  - Virtual server name to access replicas
  - Managed by WSFC
- Mirroring Endpoint
  - Used to send log records
  - Synchronization mechanism similar to Database Mirroring

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▶ Synchronization and Failover

- Availability Mode
  - Synchronous
  - Asynchronous
- Failover mode
  - Automatic
  - Manual
  - Forced

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▶ Conditions for Automatic Failover

- Primary and Secondary need
  - Synchronous Availability Mode configuration
  - Automatic Failover configuration
  - To be synchronized
- WFSC needs to reach a quorum
- Failover policies met

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▶ Flexible failover policies for Availability Groups

- Leverages FCI mechanism
  - **sp_server_diagnostic**
- Configuration in T-SQL
  - **ALTER AVAILABILITY GROUP …**
    SET (HEALTH_CHECK_TIMEOUT = 30000);
  - **ALTER AVAILABILITY GROUP …**
    SET (FAILURE_CONDITION_LEVEL = 3);

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▶ Failure condition levels for Availability Groups

- On server down
- On server unresponsive
- On critical server error
- On moderate server error
- On any qualified failure conditions

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▶ Replicas and backups

- Backup support on secondary replicas
  - Full backups with **COPY_ONLY**
  - Differential not supported!
  - Log backups
    - Form a single log-chain across all replicas
- Automated Backup Preference & Priority
  - No immediate effect
  - Backup scripts need to query UDF
    - **sys.fn_hadr_backup_is_preferred_replica('dbname')**

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▶ Read-Only Access

- Replicas roles
  - Primary: **READ_WRITE | ALL**
  - Secondary: **NO | READ_ONLY | ALL**
- New connection property
  - **ApplicationIntent**
  - Name slightly different across client APIs
- Read-Only Routing

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▶ Availability Group Listener

- Defined by
  - Virtual Network Name (VNN)
  - TCP Port
  - One or more IP addresses (static or DHCP)
- Client connection
  - Application Intent allows automatic routing
  - Support for Mirroring connection string
  - SPN registration automatic for Kerberos auth.

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▶ Topology example 1

Direct attached storage local, regional and geo secondaries

Synchronous data movement

Asynchronous data movement

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▶ Demo

- Primary Datacenter
  - Primary replica
  - Secondary replica for HA and reporting
- Secondary Datacenter
  - Secondary replica for DR

Step-by-step AG

- Cluster configuration before starting

Server Manager

File Action View Help

Server Manager (SQL1)
- Roles
- Features
  - Failover Cluster Manager
    - SQLCLUSTER.solidq.loc
      - Services and applications
      - Nodes
        - SQL1
        - SQL2
        - SQL3
      - Storage
      - Networks
        - Cluster Network 1
      - Cluster Events
- Diagnostics
- Configuration
- Storage

SQLCLUSTER.solidq.loc

Cluster SQLCLUSTER.solidq.loc

Summary of Cluster SQLCLUSTER

SQLCLUSTER has 0 applications/services and 3 nodes

Name: SQLCLUSTER.solidq.loc
Networks: Cluster Network 1
Current Host Server: SQL1
Subnets: 1 IPv4 and 0 IPv6
Quorum Configuration: Node Majority
Recent Cluster Events: Critical:2

Configure

Navigate

Cluster Core Resources

Name	Status
Cluster Name	
Name: SQLCLUSTER	Online
IP Address: 10.0.0.20	Online

More Information

Actions

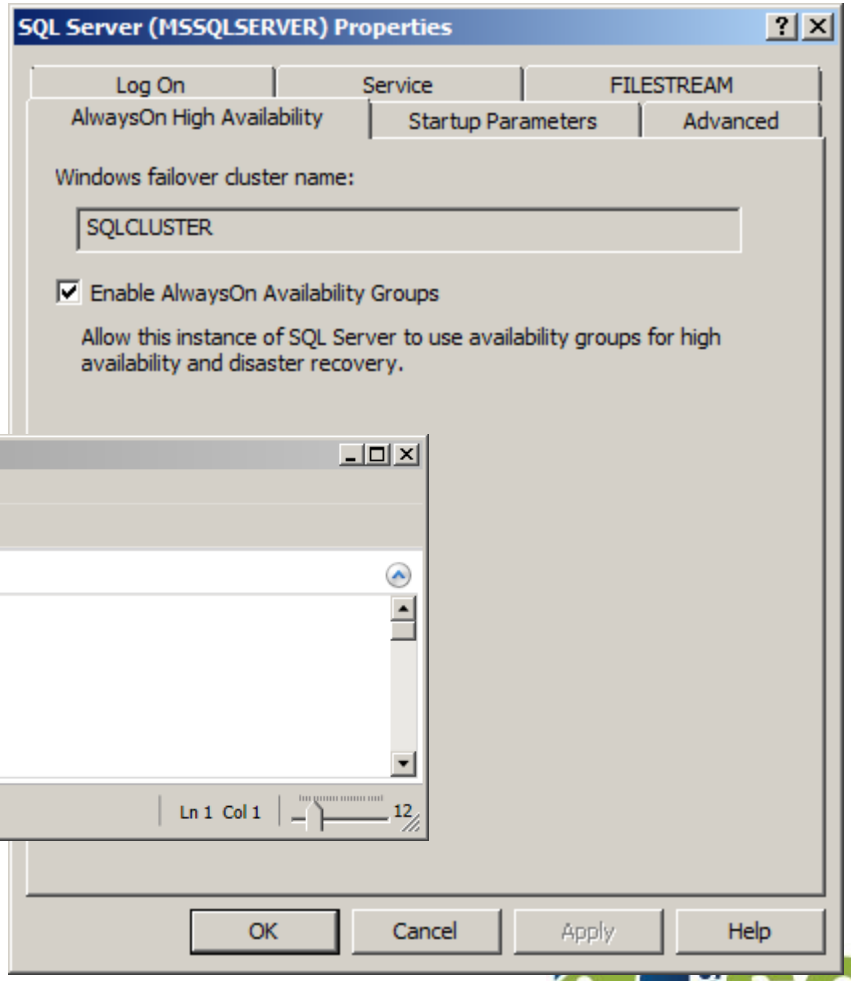
SQLCLUSTER.solidq.loc
- Configure a Service or Application...
- Validate This Cluster...
- View Validation Report
- Enable Cluster Shared Volumes...
- Add Node...
- Close Connection
- More Actions...
- View
- Refresh
- Properties
- Help

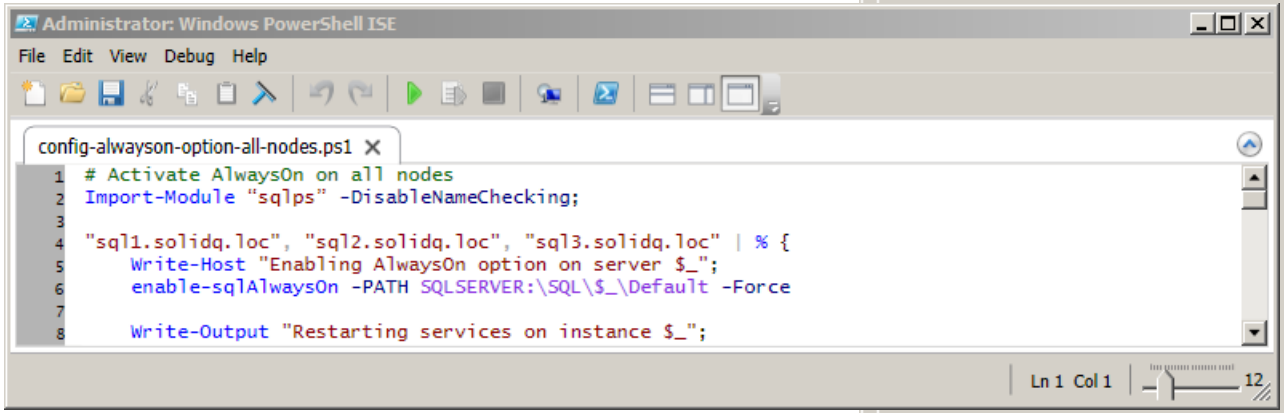
Name: SQLCLUSTER
- Bring this resource online
- Take this resource offline
- Show the critical events for this re...
- Show Dependency Report
- More Actions...
- Properties
- Help

Step-by-step AG

▶ Enable AlwaysOn

- Enable option
- Restart Services





Step-by-step AG

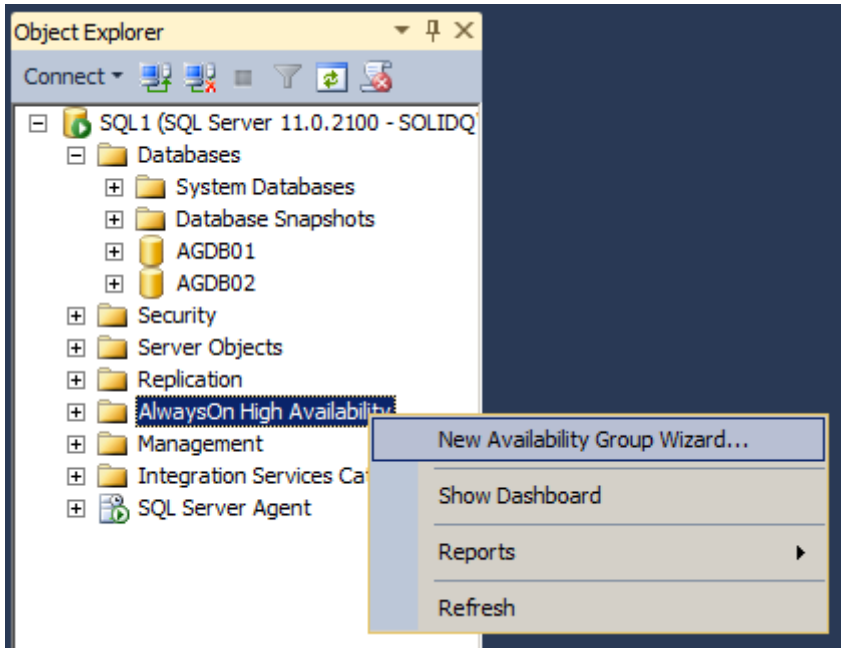
- Create sample databases

create-databases.sql - SQL1.master (SOLIDQ\Administrator (54)) - Microsoft SQL Server Management Studio (Administrator)

```
IF DB_ID('AGDB01') IS NOT NULL
    DROP DATABASE AGDB01;
GO
CREATE DATABASE AGDB01;
GO
IF DB_ID('AGDB02') IS NOT NULL
    DROP DATABASE AGDB02;
GO
CREATE DATABASE AGDB02;
GO

CREATE TABLE AGDB01.dbo.T1 (F1 int not null PRIMARY KEY);
GO
INSERT  AGDB01.dbo.T1
VALUES  (1), (2), (3), (4), (5);
GO
CREATE TABLE AGDB02.dbo.T2 (F1 int not null PRIMARY KEY);
GO
INSERT  AGDB02.dbo.T2
VALUES  (1), (2), (3), (4), (5);
GO
```

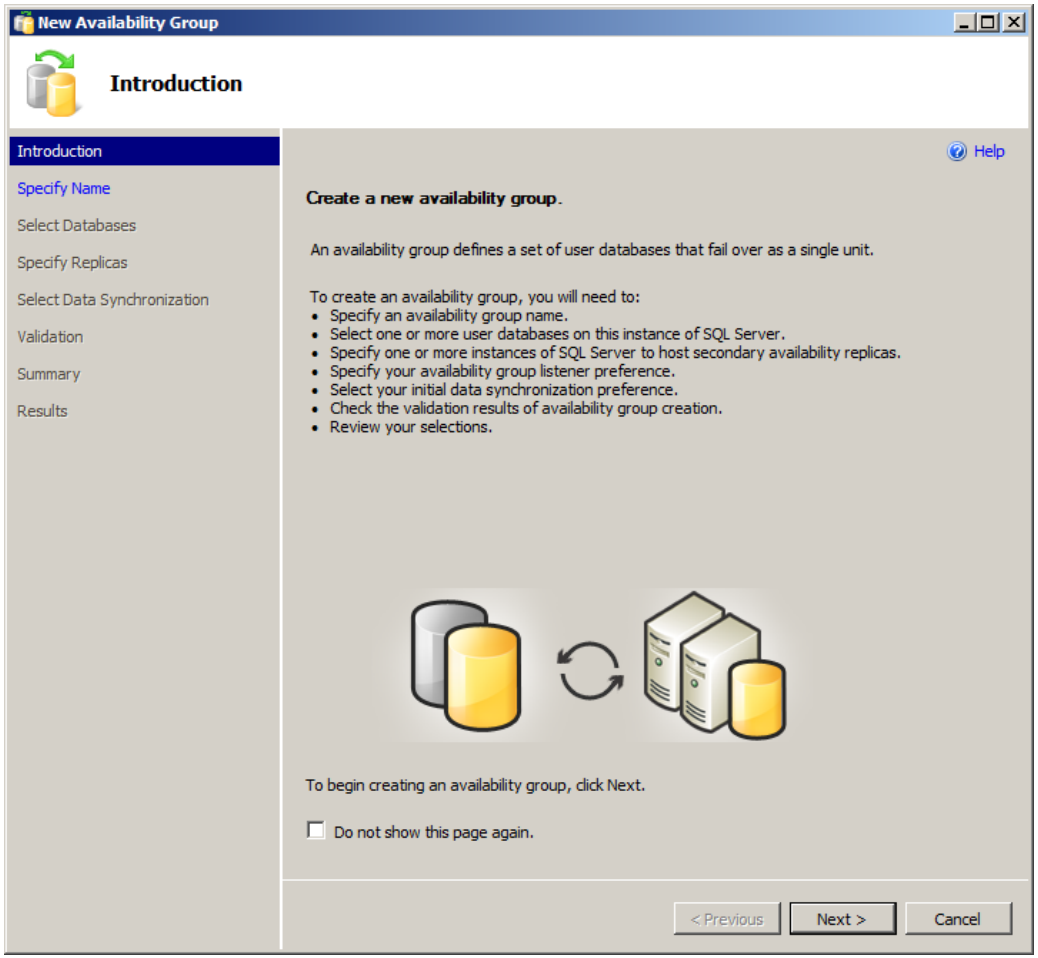
Step-by-step AG

▶ Select Availability Groups Wizard



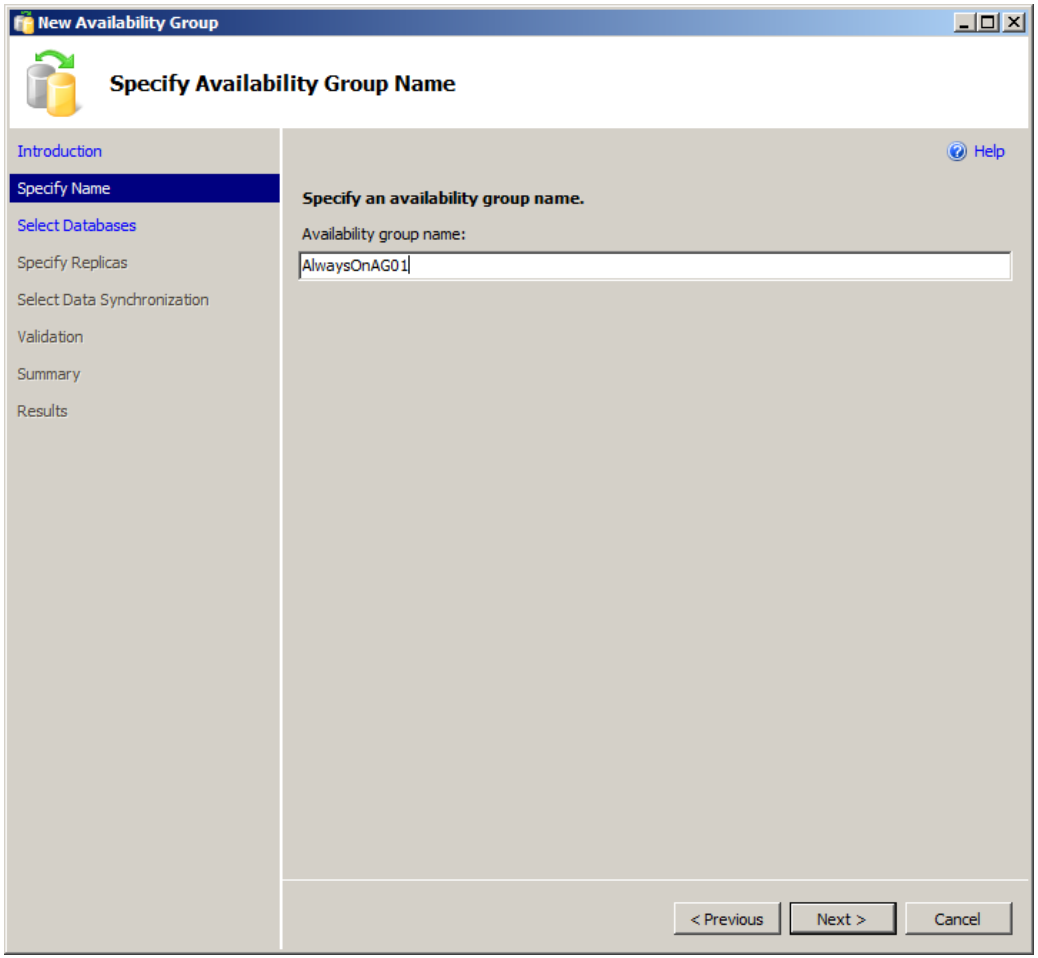
Step-by-step AG

▶ AG Wizard: Start



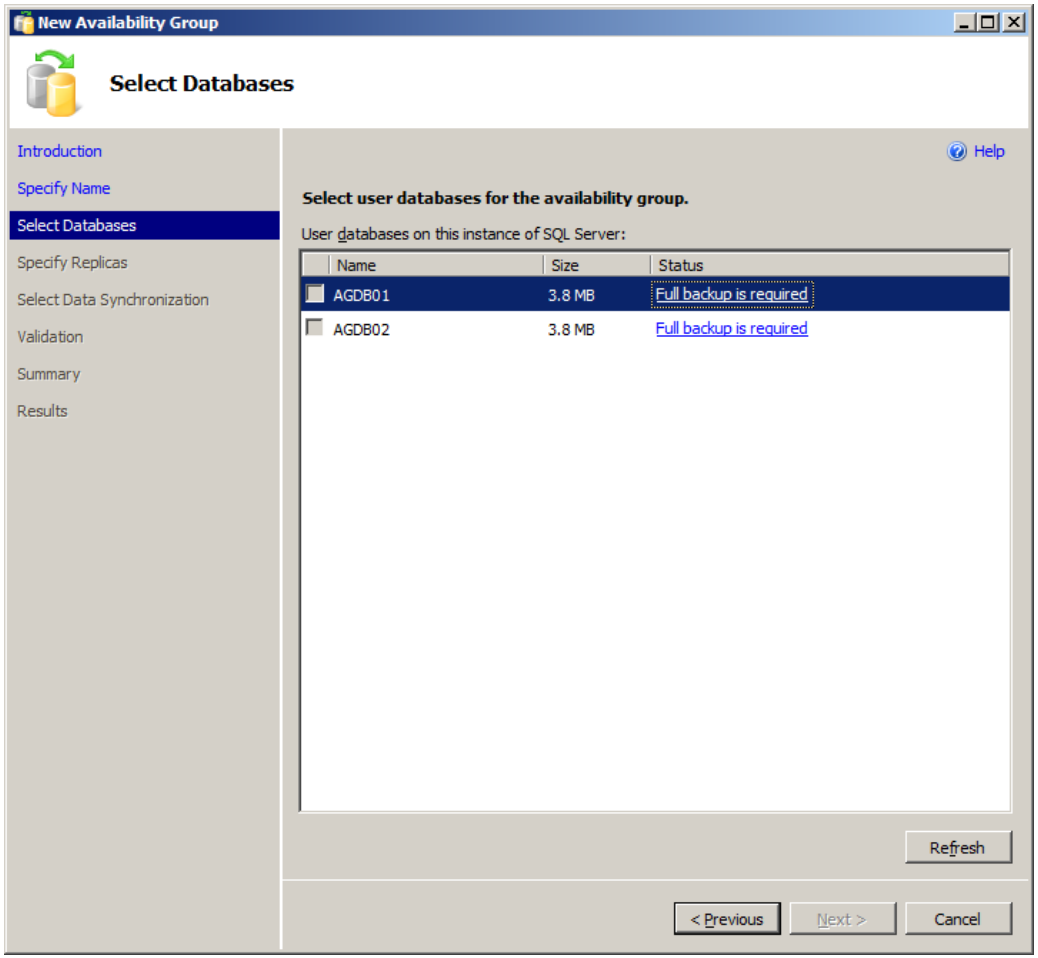
Step-by-step AG

▶ AG Wizard: Assign Name to Group



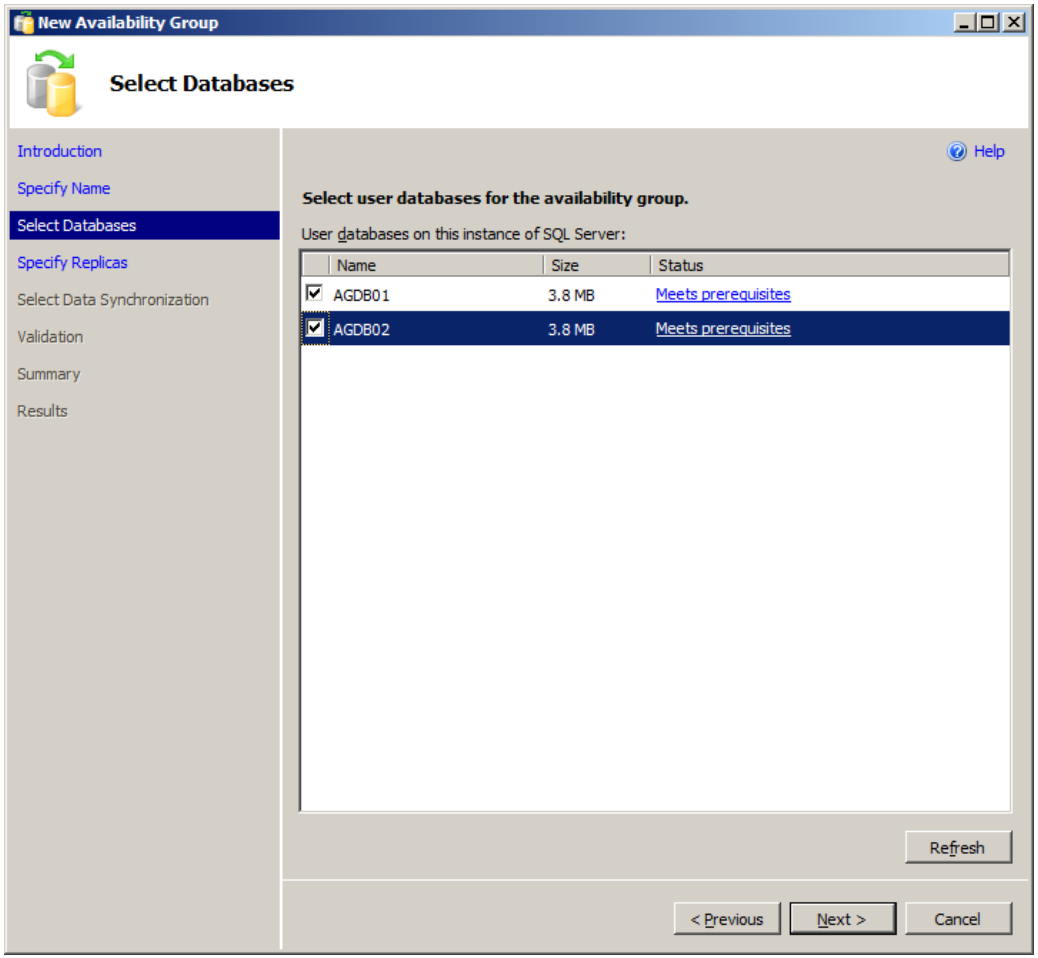
Step-by-step AG

- AG Wizard: full backup required to choose a database



Step-by-step AG

▶ AG Wizard: Select databases



Step-by-step AG

▶ AG Wizard: configure Replicas

New Availability Group

Specify Replicas

- Introduction
- Specify Name
- Select Databases
- Specify Replicas
- Select Data Synchronization
- Validation
- Summary
- Results

Help

Specify an instance of SQL Server to host a secondary replica.

Replicas | Endpoints | Backup Preferences | Listener

Availability Replicas:

Server Instance	Initial Role	Automatic Failover (Up to 2)	Synchronous Commit (Up to 3)	Readable Secondary
SQL1	Primary	☑	☑	No
SQL2	Secondary	☑	☑	No
SQL3	Secondary	☐	☐	No

Add Replica... | Remove Replica

Summary for the replica hosted by SQL2

Replica mode: Synchronous commit with automatic failover
This replica will use synchronous-commit availability mode and support both automatic failover and manual failover.

Readable secondary: No
In the secondary role, this availability replica will not allow any connections.

< Previous | Next > | Cancel

Step-by-step AG

▶ AG Wizard: configure Endpoints

New Availability Group

Specify Replicas

- Introduction
- Specify Name
- Select Databases
- Specify Replicas
- Select Data Synchronization
- Validation
- Summary
- Results

Help

Specify an instance of SQL Server to host a secondary replica.

Replicas | Endpoints | Backup Preferences | Listener

Endpoint values:

Server Name	Endpoint URL	Port Number	Endpoint Name	Encrypt Data	SQL Server Service Account
SQL1	TCP://SQL1.solidq.loc:5022	5022	Hadr_endpoint	☑	SOLIDQ\SQLService
SQL2	TCP://SQL2.solidq.loc:5022	5022	Hadr_endpoint	☑	SOLIDQ\SQLService
SQL3	TCP://SQL3.solidq.loc:5022	5022	Hadr_endpoint	☑	SOLIDQ\SQLService

Refresh

< Previous | Next > | Cancel

Step-by-step AG

▶ AG Wizard: configure Backup preferences

New Availability Group

Specify Replicas

Introduction
Specify Name
Select Databases
Specify Replicas
Select Data Synchronization
Validation
Summary
Results

Help

Specify an instance of SQL Server to host a secondary replica.

Replicas | Endpoints | Backup Preferences | Listener

Where should backups occur?

- **Prefer Secondary**
  Automated backups for this availability group should occur on a secondary replica. If there is no secondary replica available, backups will be performed on the primary replica.
- **Secondary only**
  All automated backups for this availability group must occur on a secondary replica.
- **Primary**
  All automated backups for this availability group must occur on the current primary replica.
- **Any Replica**
  Backups can occur on any replica in the availability group.

Replica backup priorities:

Server Instance	Backup Priority (Lowest=1, Highest=100)	Exclude Replica
SQL1	50	☐
SQL2	100	☐
SQL3	50	☑

< Previous | Next > | Cancel

Step-by-step AG

▶ AG Wizard: configure Listener

New Availability Group

Specify Replicas

Introduction
Specify Name
Select Databases
Specify Replicas
Select Data Synchronization
Validation
Summary
Results

Help

Specify an instance of SQL Server to host a secondary replica.

Replicas | Endpoints | Backup Preferences | Listener

Specify your preference for an availability group listener that will provide a client connection point:

○ **Do not create an availability group listener now**
You can create the listener later using the Add Availability Group Listener dialog.

◉ **Create an availability group listener**
Specify your listener preferences for this availability group.

Listener DNS Name: AGVNN
Port: 1433
Network Mode: Static IP

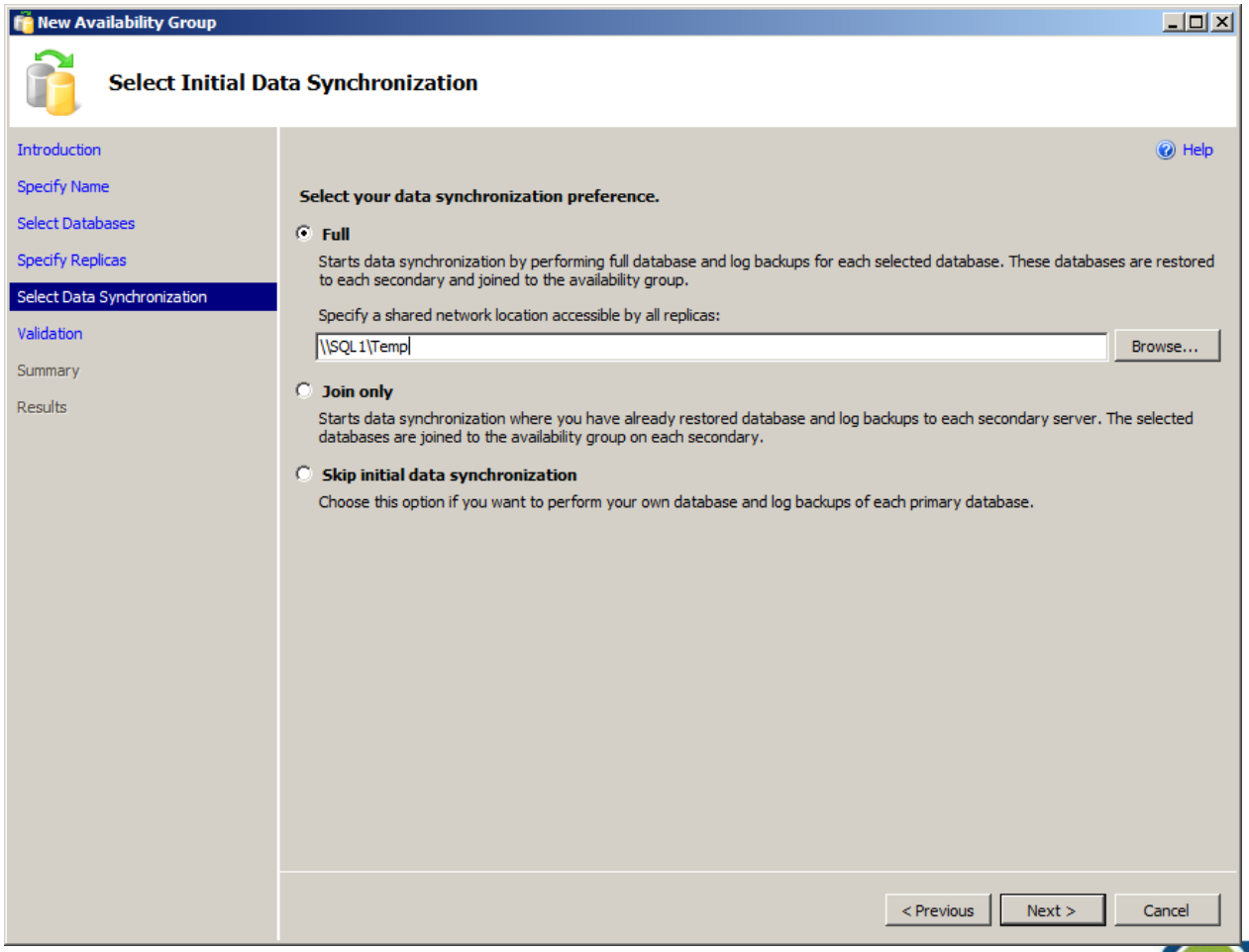
Subnet	IP Address
10.0.0.0/24	10.0.0.30

Add... Remove

< Previous Next > Cancel

Step-by-step AG

▶ AG Wizard: configure data synchronization



Step-by-step AG

▶ AG Wizard: Validation

New Availability Group

Validation

- Introduction
- Specify Name
- Select Databases
- Specify Replicas
- Select Data Synchronization
- Validation
- Summary
- Results

Help

Results of availability group validation.

	Name	Result
✓	Checking whether the endpoint is encrypted using a compatible algorithm	Success
✓	Checking shared network location	Success
✓	Checking for free disk space on the server instance that hosts secondary replica SQL2	Success
✓	Checking if the selected databases already exist on the server instance that hosts secondary replica SQL2	Success
✓	Checking for compatibility of the database file locations on the server instance that hosts secondary replica SQL2	Success
✓	Checking for the existence of the database files on the server instance that hosts secondary replica SQL2	Success
✓	Checking for free disk space on the server instance that hosts secondary replica SQL3	Success
✓	Checking if the selected databases already exist on the server instance that hosts secondary replica SQL3	Success
✓	Checking for compatibility of the database file locations on the server instance that hosts secondary replica SQL3	Success
✓	Checking for the existence of the database files on the server instance that hosts secondary replica SQL3	Success
✓	Checking the listener configuration	Success

Re-run Validation

< Previous | Next > | Cancel

Step-by-step AG

▶ AG Wizard: Summary

New Availability Group

Summary

- Introduction
- Specify Name
- Select Databases
- Specify Replicas
- Select Data Synchronization
- Validation
- Summary
- Results

Help

Verify the choices made in this wizard.

Click Finish to perform the following actions:

- Availability Group: AlwaysOnAG01
  - Primary replica: SQL1
  - Availability Group Listener: AGVNN
  - Automated backup preference: Secondary
  - Databases
    - AGDB01 (3.8 MB)
    - AGDB02 (3.8 MB)
  - Initial data synchronization: Full
  - Backup location: \\SQL1\Temp
  - Replicas
    - Server instance name: SQL1
      - Role: Primary
      - Replica mode: Synchronous commit with automatic failover
      - Readable secondary: No
      - Endpoint: Hadr_endpoint
        - URL: TCP://SQL1.solidq.loc:5022
        - Encrypted: Yes
        - Service account: SOLIDQ\SQLService
      - Automated backup priority: 50
      - New Logins:
        - SOLIDQ\SQLService
    - Server instance name: SQL2
      - Role: Secondary
      - Replica mode: Synchronous commit with automatic failover

Script

< Previous | Finish | Cancel

Step-by-step AG

▶ AG Wizard: Progress

New Availability Group

Progress

- Introduction
- Specify Name
- Select Databases
- Specify Replicas
- Select Data Synchronization
- Validation
- Summary
- Results

Help

Starting the 'AlwaysOn_health' XEvent session on 'SQL1'.

Name	Status
Creating logins on 'SQL1'.	Success
Creating logins on 'SQL2'.	Success
Creating logins on 'SQL3'.	Success
Configuring endpoints.	Success
Starting the 'AlwaysOn_health' XEvent session on 'SQL1'.	In Progress
Starting the 'AlwaysOn_health' XEvent session on 'SQL2'.	Not started
Starting the 'AlwaysOn_health' XEvent session on 'SQL3'.	Not started
Creating availability group 'AlwaysOnAG01'.	Not started
Waiting for availability group 'AlwaysOnAG01' to come online.	Not started
Create Availability Group Listener 'AGVNN'.	Not started
Joining secondary replicas to availability group 'AlwaysOnAG01'.	Not started
Validating WSFC quorum vote configuration.	Not started
Creating a full backup for 'AGDB01'.	Not started
Restoring 'AGDB01' on 'SQL2'.	Not started
Restoring 'AGDB01' on 'SQL3'.	Not started

Fewer details

< Previous | Next > | Cancel

Step-by-step AG

▶ AG Wizard: Results

New Availability Group

Results

- Introduction
- Specify Name
- Select Databases
- Specify Replicas
- Select Data Synchronization
- Validation
- Summary
- Results

Help

The wizard finished with warnings.

Summary:

Name	Result
Creating logins on 'SQL1'.	Success
Creating logins on 'SQL2'.	Success
Creating logins on 'SQL3'.	Success
Configuring endpoints.	Success
Starting the 'AlwaysOn_health' XEvent session on 'SQL1'.	Success
Starting the 'AlwaysOn_health' XEvent session on 'SQL2'.	Success
Starting the 'AlwaysOn_health' XEvent session on 'SQL3'.	Success
Creating availability group 'AlwaysOnAG01'.	Success
Waiting for availability group 'AlwaysOnAG01' to come online.	Success
Create Availability Group Listener 'AGVNN'.	Success
Joining secondary replicas to availability group 'AlwaysOnAG01'.	Success
Validating WSFC quorum vote configuration.	Warning
Creating a full backup for 'AGDB01'.	Success
Restoring 'AGDB01' on 'SQL2'.	Success
Restoring 'AGDB01' on 'SQL3'.	Success
Backing up log for 'AGDB01'.	Success

< Previous | Next > | Close

Step-by-step AG

▶ AG Dashboard

AlwaysOnAG01:SQL1 - Microsoft SQL Server Management Studio (Administrator)

File Edit View Project Debug Tools Window Help

New Query

Object Explorer

Connect

SQL1 (SQL Server 11.0.2100 - SOLIDQ\Adr
- Databases
  - System Databases
  - Database Snapshots
  - AGDB01
  - AGDB02
- Security
- Server Objects
- Replication
- AlwaysOn High Availability
  - Availability Groups
    - AlwaysOnAG01 (Primary)
      - Availability Replicas
        - SQL1 (Primary)
        - SQL2 (Secondar
        - SQL3 (Secondar
      - Availability Databas
        - AGDB01
        - AGDB02
      - Availability Group Li
        - AGVNN
- Management
- Integration Services Catalogs
- SQL Server Agent

Add Database...
Add Replica...
Add Listener...
Show Dashboard
Failover...
Start PowerShell
Reports
Delete...
Refresh
Properties

AlwaysOnAG01:SQL1 | Dashboard: SQL1

AlwaysOnAG01: hosted by SQL1 (Replica role: Primary)

Last updated: 3/31/2012 2:26:57 AM

Auto refresh: on

Start Failover Wizard
View AlwaysOn Health Events
View Cluster Quorum Information

Availability group state: Healthy
Primary instance: SQL1
Failover mode: Automatic
Cluster state: SQLCLUSTER (Normal Quorum)

Availability replica:

Name	Role	Failover Mode	Synchronization State	Issues
SQL1	Primary	Automatic	Synchronized	
SQL2	Secondary	Automatic	Synchronized	
SQL3	Secondary	Manual	Synchronizing	

Replica	Synchronization State	Failover Readin...	Issues
SQL1	Synchronized	No Data Loss	
SQL1	Synchronized	No Data Loss	
SQL2	Synchronized	No Data Loss	
SQL2	Synchronized	No Data Loss	
AGDB01 SQL3	Synchronizing	Data Loss	
AGDB02 SQL3	Synchronizing	Data Loss	

Ready

Step-by-step AG

- Cluster configuration of AG after installation

Server Manager

File Action View Help

Server Manager (SQL1)
- Roles
- Features
- Failover Cluster Manager
  - SQLCLUSTER.solidq.loc
    - Services and application
      - AlwaysOnAG01
    - Nodes
      - SQL1
      - SQL2
      - SQL3
    - Storage
    - Networks
      - Cluster Network 1
    - Cluster Events
- Diagnostics
- Configuration
- Storage

AlwaysOnAG01

AlwaysOnAG01 — Recent Cluster Events: Critical:11, Error:1

Summary of AlwaysOnAG01

Status: Online
Auto Start: Yes
Alerts: <none>
Preferred Owners: SQL2, SQL1
Current Owner: SQL2

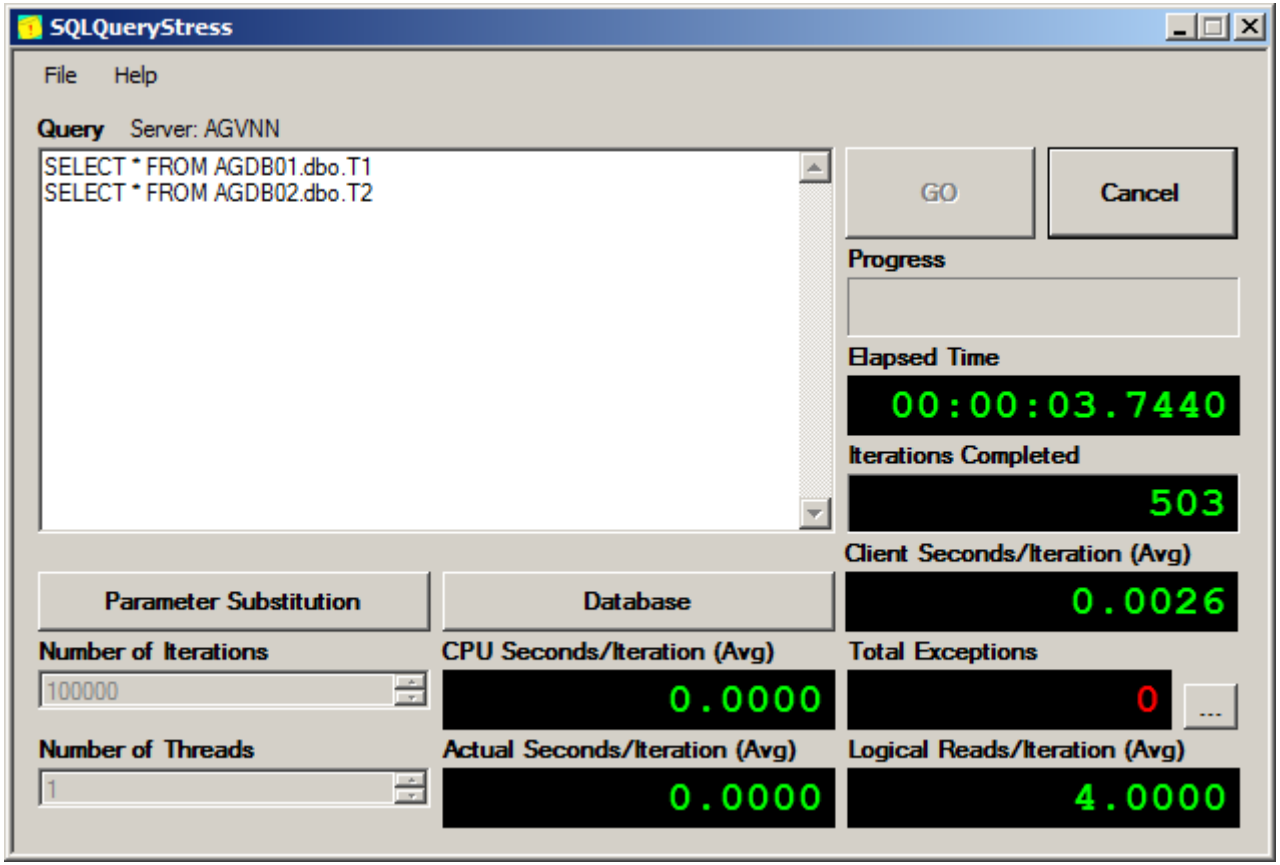
Name	Status
Server Name	
Name: AGVNN	Online
IP Address: 10.0.0.30	Online
Other Resources	
AlwaysOnAG01	Online

Actions

AlwaysOnAG01
- Bring this service or application online
- Take this service or application offline
- Move this service or application to anot...
- Manage shares and storage
- Add a shared folder
- Show the critical events for this applica...
- Add storage
- Add a resource
- Disable auto start
- Show Dependency Report
- View
- Delete
- Rename
- Refresh
- Properties
- Help

Step-by-step AG

- Simulate a failure: start a workload



Step-by-step AG

- Simulate a failure: start a manual Failover

AlwaysOnAG01:SQL1 - Microsoft SQL Server Management Studio (Administrator)

File Edit View Project Debug Tools Window Help

New Query

Object Explorer

Connect

SQL1 (SQL Server 11.0.2100 - SOLIDQ\Adr
- Databases
  - System Databases
  - Database Snapshots
  - AGDB01
  - AGDB02
- Security
- Server Objects
- Replication
- AlwaysOn High Availability
  - Availability Groups
    - AlwaysOnAG01 (Primary)
      - Availability Replicas
        - SQL1 (Primary)
        - SQL2 (Seconda
        - SQL3 (Seconda
      - Availability Databa
        - AGDB01
        - AGDB02
      - Availability Group L
        - AGVNN
- Management
- Integration Services Catalogs
- SQL Server Agent

Add Database...
Add Replica...
Add Listener...
Show Dashboard
Failover...
Start PowerShell
Reports
Delete...
Refresh
Properties

AlwaysOnAG01:SQL1 | Dashboard: SQL1

AlwaysOnAG01: hosted by SQL1 (Replica role: Primary)

Last updated: 3/31/2012 2:16:54 PM

Auto refresh: on

Start Failover Wizard
View AlwaysOn Health Events
View Cluster Quorum Information

Availability group state: Healthy
Primary instance: SQL1
Failover mode: Automatic
Cluster state: SQLCLUSTER (Normal Quorum)

Availability replica:

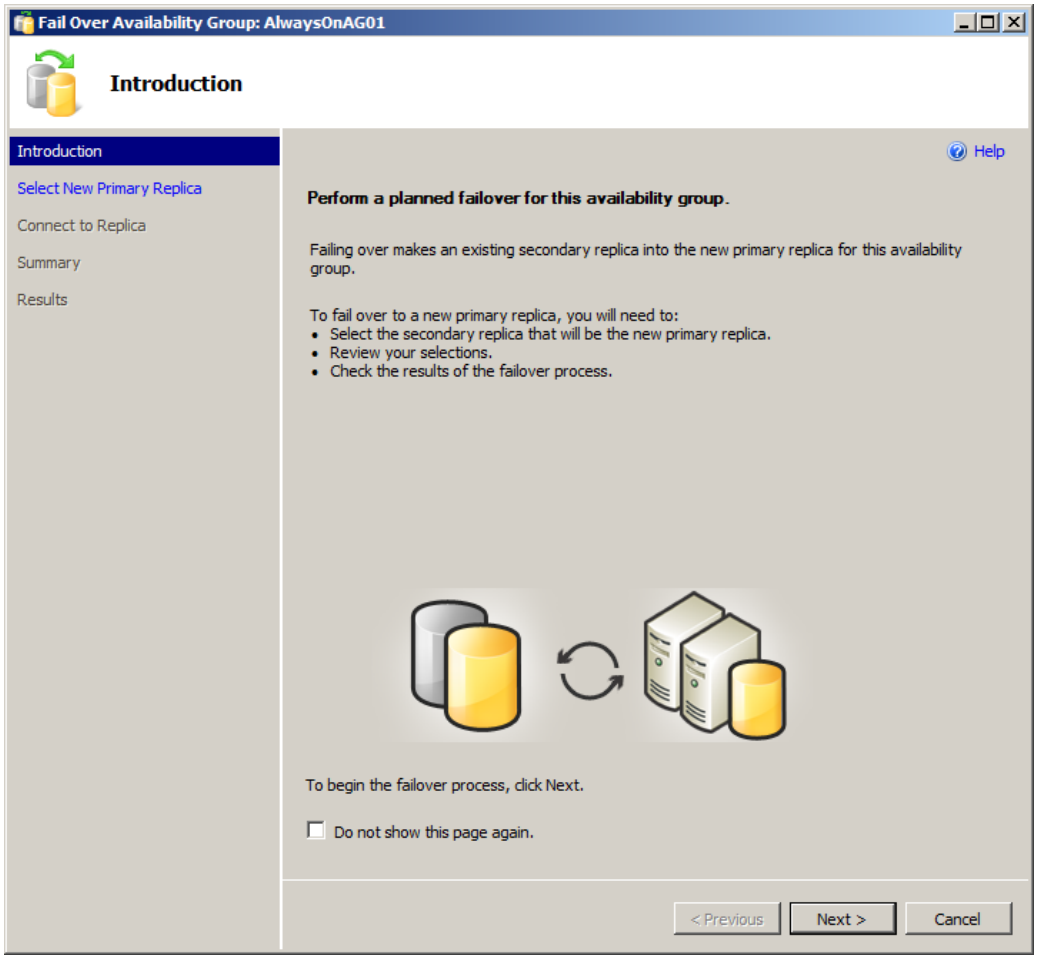
Name	Role	Failover Mode	Synchronization State	Issues
SQL1	Primary	Automatic	Synchronized	
SQL2	Second...	Automatic	Synchronized	
	Second...	Manual	Synchronizing	

Replica	Synchronization State	Failover Readin...	Issues
SQL1	Synchronized	No Data Loss	
SQL1	Synchronized	No Data Loss	
SQL2	Synchronized	No Data Loss	
SQL2	Synchronized	No Data Loss	
AGDB01 SQL3	Synchronizing	Data Loss	
AGDB02 SQL3	Synchronizing	Data Loss	

Ready

Step-by-step AG

▶ Failover Wizard: Introduction



Step-by-step AG

▶ Failover Wizard: select new Primary Replica

Fail Over Availability Group: AlwaysOnAG01

Select New Primary Replica

Introduction
Select New Primary Replica
Connect to Replica
Summary
Results

Help

Select the new primary replica for this availability group.

Current Primary Replica: SQL1
Primary Replica Status: Synchronous commit and Online
Quorum Status: Normal Quorum

Choose new primary replica:

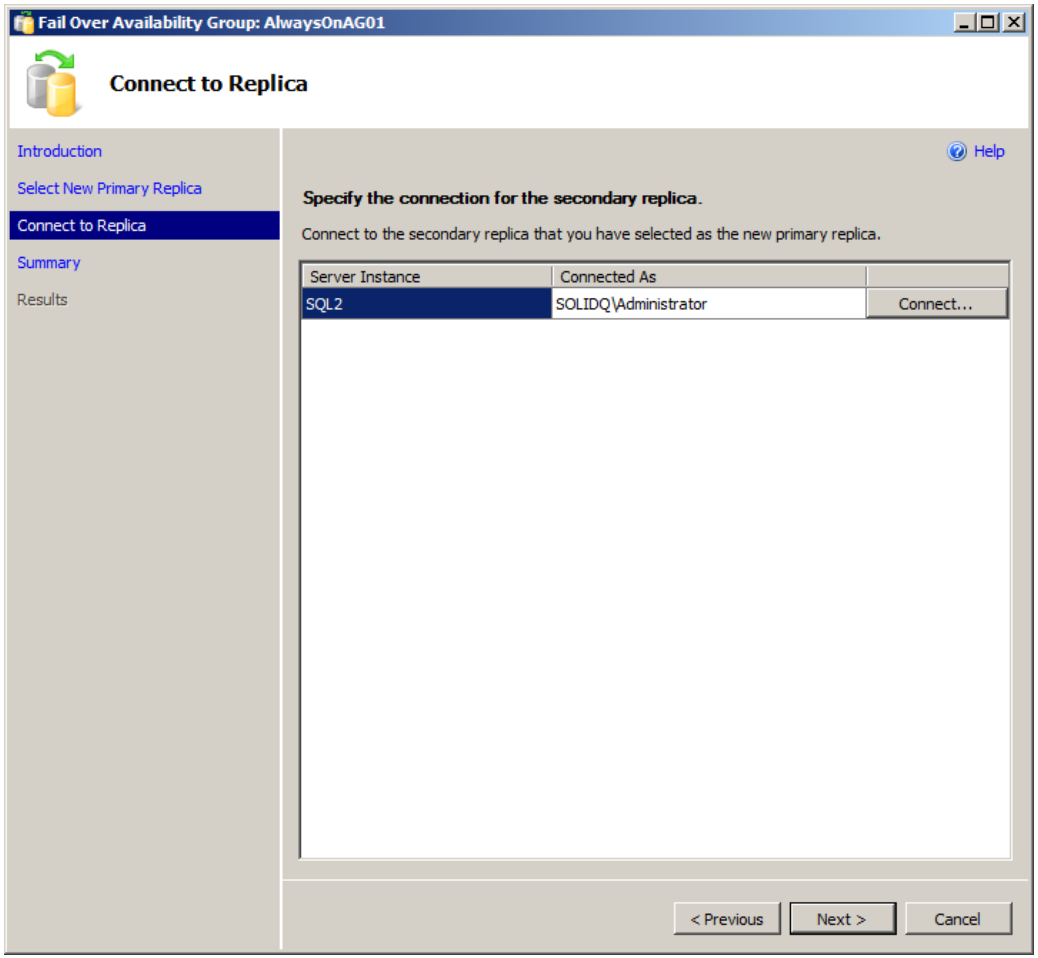
		Server Instance	Availability Mode	Failover M...	Failover Readiness
✓	☑	SQL2	Synchronous commit	Automatic	No data loss, Warnings(1)
⚠	☐	SQL3	Asynchronous commit	Manual	Data loss, Warnings(3)

Refresh

< Previous | Next > | Cancel

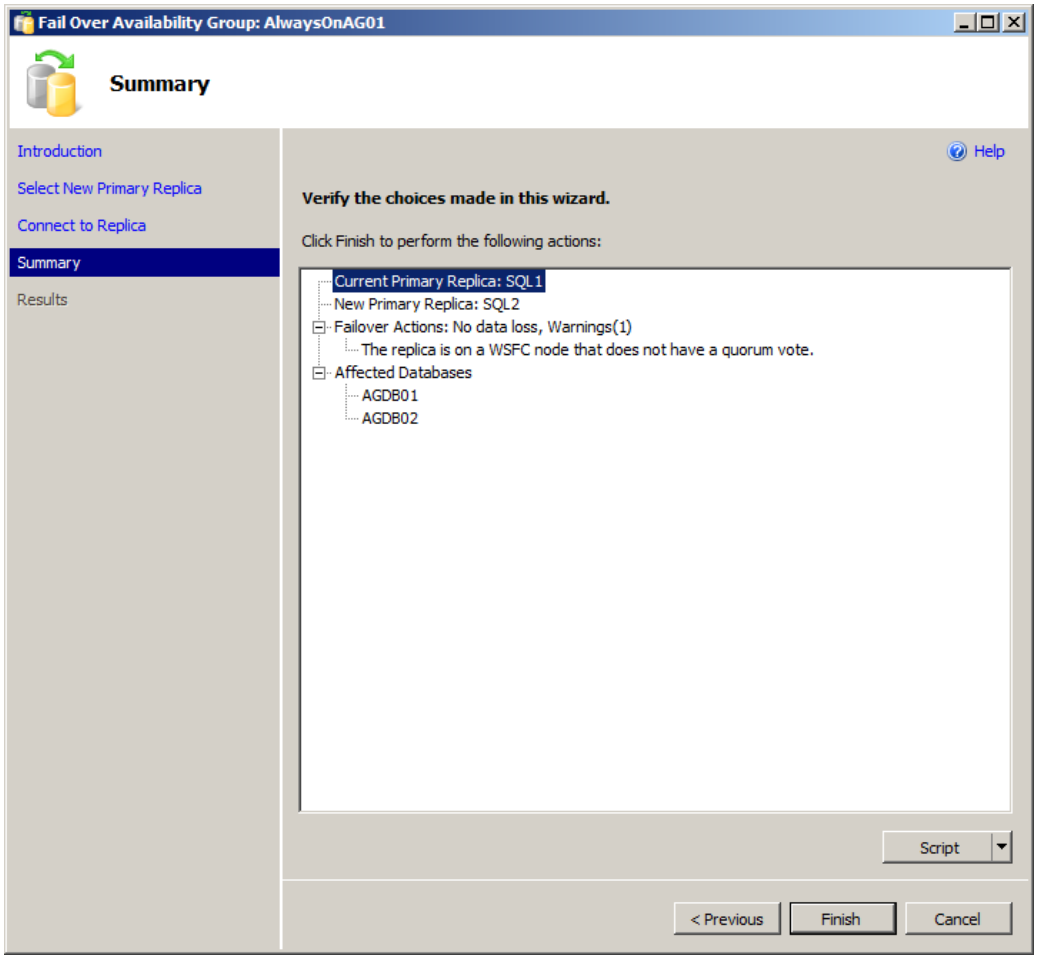
Step-by-step AG

▶ Failover Wizard: connect to Secondary Replica



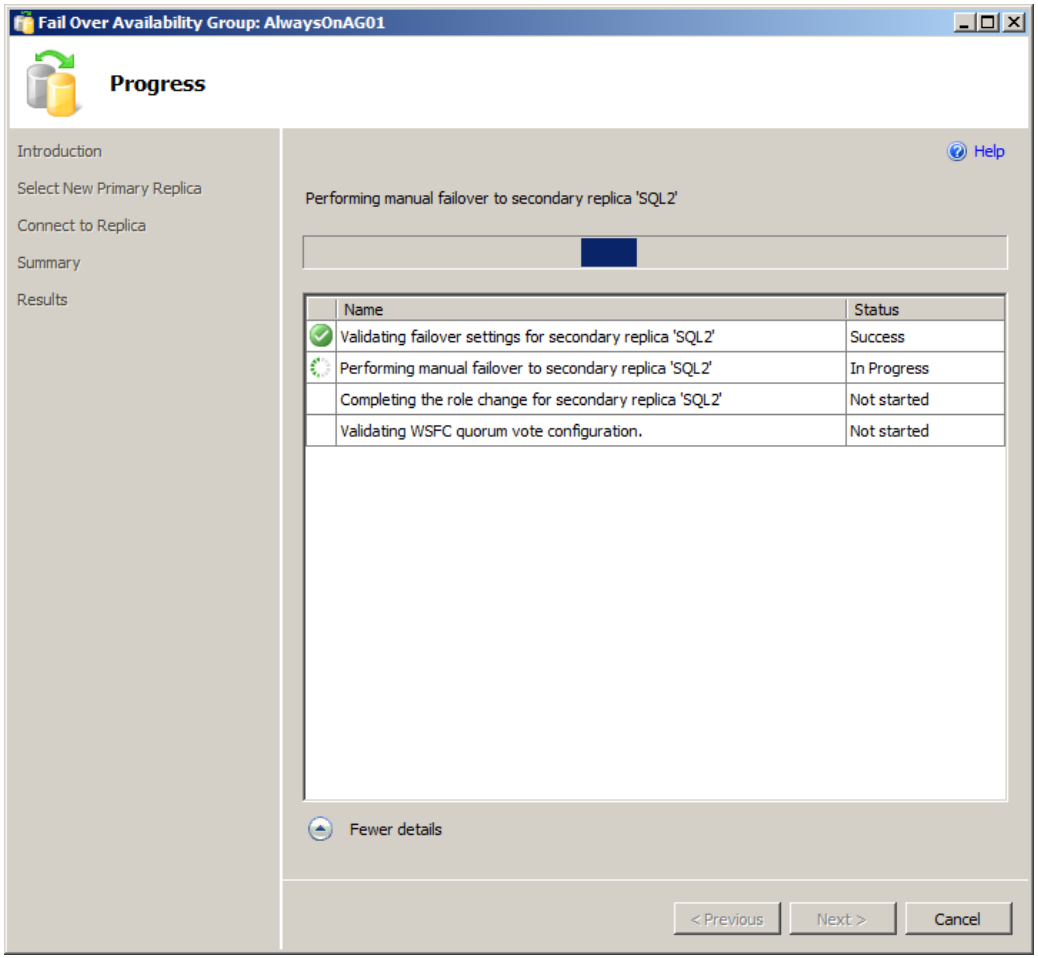
Step-by-step AG

▶ Failover Wizard: Summary



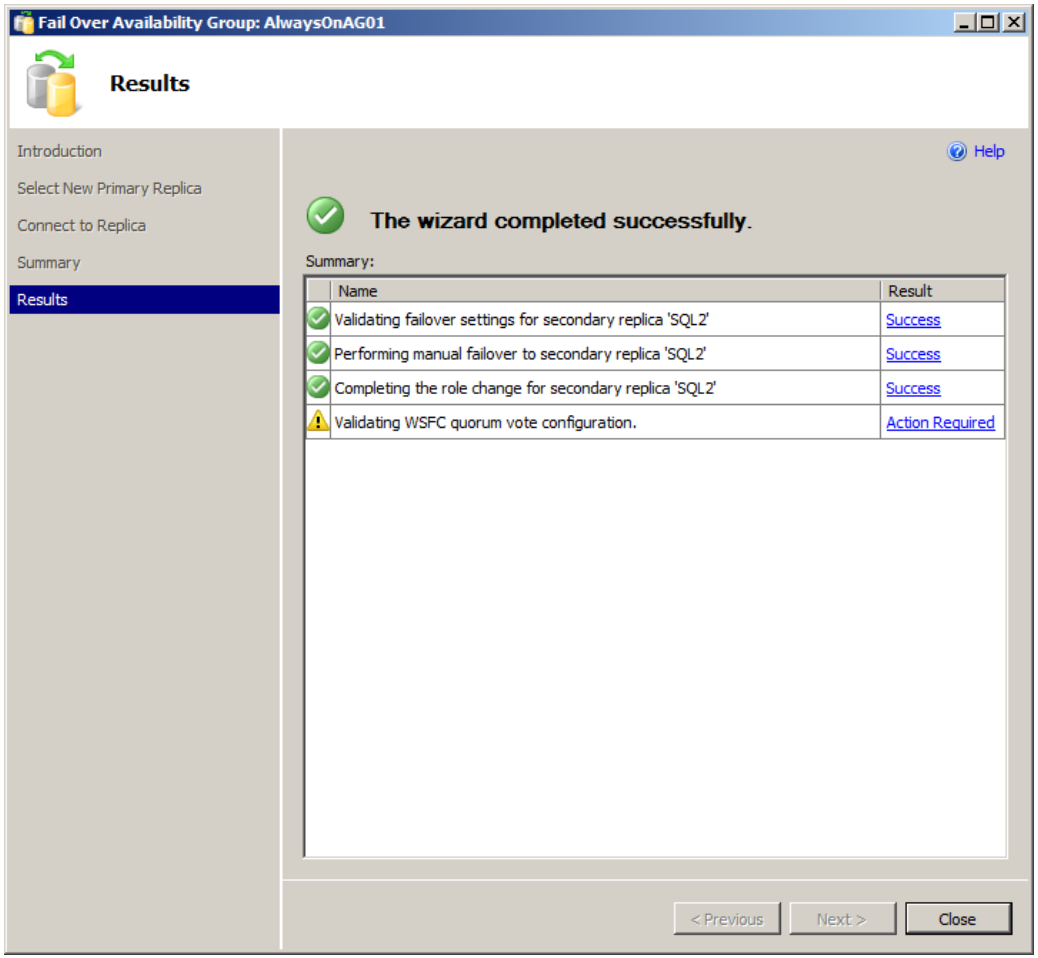
Step-by-step AG

▶ Failover Wizard: Progress



Step-by-step AG

▶ Failover Wizard: Complete



Step-by-step AG

- Simulate a failure: Dashboard on new Secondary Replica

AlwaysOnAG01:SQL1 - Microsoft SQL Server Management Studio (Administrator)

File Edit View Project Debug Tools Window Help

New Query

Object Explorer

Connect

SQL1 (SQL Server 11.0.2100 - SOLIDQ\Adr
- Databases
  - System Databases
  - Database Snapshots
  - AGDB01
  - AGDB02
- Security
- Server Objects
- Replication
- AlwaysOn High Availability
  - Availability Groups
    - AlwaysOnAG01 (Secondary)
      - Availability Replicas
        - SQL1 (Secondary)
        - SQL2
        - SQL3
      - Availability Databases
        - AGDB01
        - AGDB02
      - Availability Group Listeners
        - AGVNN
- Management
- Integration Services Catalogs
- SQL Server Agent

AlwaysOnAG01:SQL1 | Dashboard: SQL1

AlwaysOnAG01: hosted by SQL1 (Replica role: Secondary)

Last updated: 3/31/2012 2:20:54 PM

Auto refresh: on

Availability group state: No data available on secondary replicas.

Primary instance: SQL2

Failover mode: Automatic

Cluster state: SQLCLUSTER (Normal Quorum)

Start Failover Wizard

View AlwaysOn Health Events

View Cluster Quorum Information

Availability replica:

Name	Role	Failover Mode	Synchronization State	Issues
SQL1	Second...	Automatic	Synchronized	

Group by

Name	Replica	Synchronization State	Failover Readin...	Issues
SQL1				
AGDB01	SQL1	Synchronized	No Data Loss	
AGDB02	SQL1	Synchronized	No Data Loss	

Ready

Step-by-step AG

- Simulate a failure: Dashboard on new Primary Replica

AlwaysOnAG01:SQL2 - Microsoft SQL Server Management Studio (Administrator)

File Edit View Project Debug Tools Window Help

New Query

Object Explorer

Connect

- SQL1 (SQL Server 11.0.2100 - SOLIDQ\Adr
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01
    - AGDB02
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Secondary)
        - Availability Replicas
          - SQL1 (Secondary)
          - SQL2
          - SQL3
        - Availability Databases
          - AGDB01
          - AGDB02
        - Availability Group Listeners
          - AGVNN
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

AlwaysOnAG01:SQL2 | AlwaysOnAG01:SQL1 | Dashboard: SQL1

AlwaysOnAG01: hosted by SQL2 (Replica role: Primary)

Last updated: 3/31/2012 2:21:41 PM

Auto refresh: on

Start Failover Wizard

View AlwaysOn Health Events

View Cluster Quorum Information

Availability group state: Healthy

Primary instance: SQL2

Failover mode: Automatic

Cluster state: SQLCLUSTER (Normal Quorum)

Availability replica:

Name	Role	Failover Mode	Synchronization State	Issues
SQL1	Second...	Automatic	Synchronized	
SQL2	Primary	Automatic	Synchronized	
SQL3	Second...	Manual	Synchronizing	

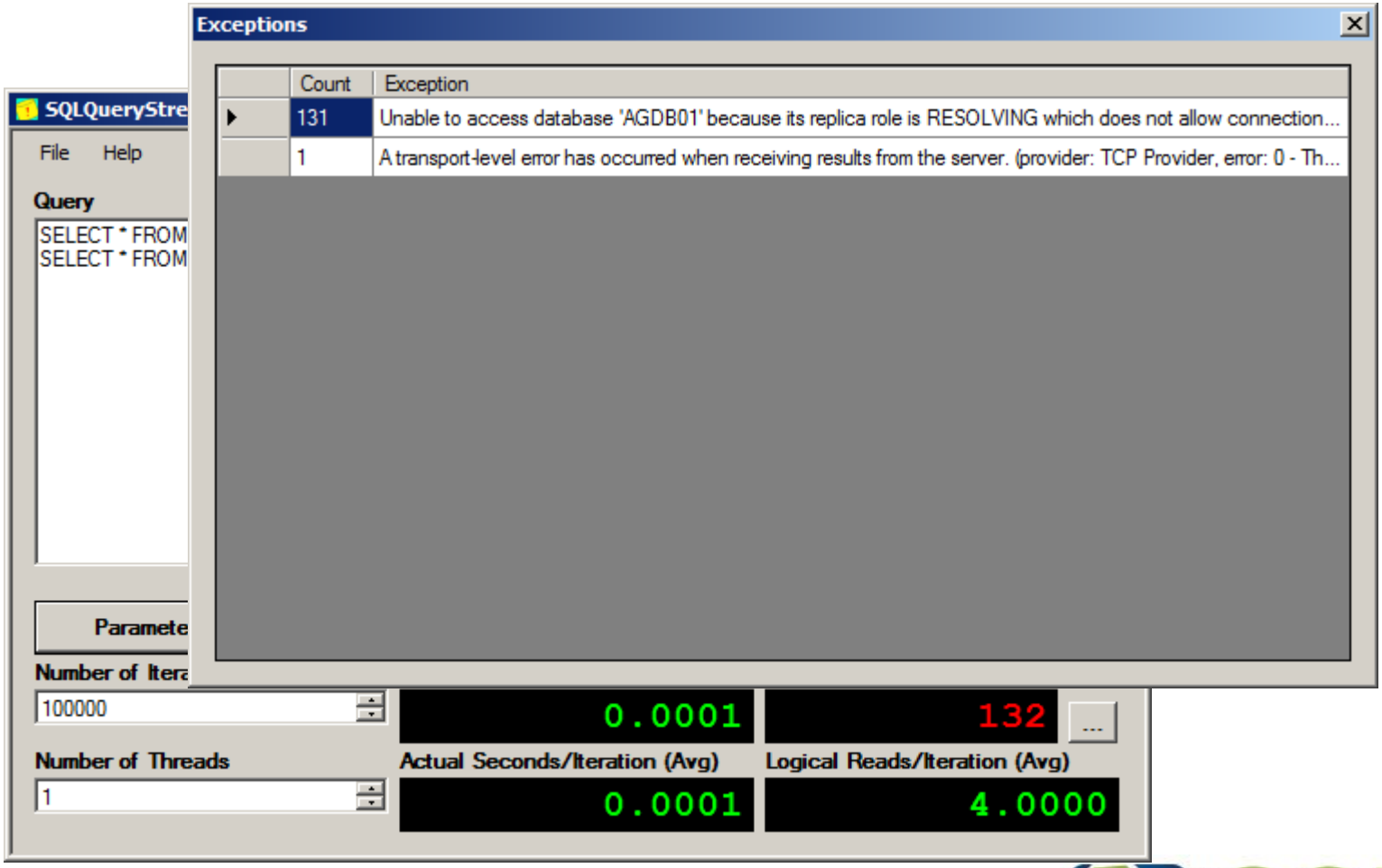
Group by

Name	Replica	Synchronization State	Failover Readin...	Issues
SQL1				
AGDB01	SQL1	Synchronized	No Data Loss	
AGDB02	SQL1	Synchronized	No Data Loss	
SQL2				
AGDB01	SQL2	Synchronized	No Data Loss	
AGDB02	SQL2	Synchronized	No Data Loss	
SQL3				
AGDB01	SQL3	Synchronizing	Data Loss	
AGDB02	SQL3	Synchronizing	Data Loss	

Ready

Step-by-step AG

- Simulate a failure: clients exceptions



Step-by-step AG

- Accessing AG: using the Listener's VNN

simple-queries.sql - AGVNN.master (SOLIDQ\Administrator (59)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- AGVNN (SQL Server 11.0.2100 - S
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (59))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL1

(1 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)
```

100 %

Query executed successfully. | AGVNN (11.0 RTM) | SOLIDQ\Administrator (59) | master | 00:00:00 | 11 rows

Item(s) Saved | Ln 5 | Col 3 | Ch 3 | INS

Step-by-step AG

▶ Accessing AG: connecting directly to Secondary Replica

Step-by-step AG

▶ Accessing AG: Secondary Replica not accessible

simple-queries.sql - SQL2.master (SOLIDQ\Administrator (52)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- AGVNN (SQL Server 11.0.2100 - S
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (52))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL2

(1 row(s) affected)

Msg 976, Level 14, State 1, Line 1
The target database, 'AGDB01', is participating in an availability group and is currently not accessible
for queries. Either data movement is suspended or the availability replica is not enabled for read access.
To allow read-only access to this and other databases in the availability group, enable read access to
one or more secondary availability replicas in the group.

For more information, see the ALTER AVAILABILITY GROUP statement in SQL Server Books Online.
```

100 %

Query completed with errors. | SQL2 (11.0 RTM) | SOLIDQ\Administrator (52) | master | 00:00:00 | 1 rows

Item(s) Saved | Ln 14 | Col 1 | Ch 1 | INS

Step-by-step AG

▶ Accessing AG: configure readable Secondary Replica

Availability Group Properties - AlwaysOnAG01

Select a page
- General
- Backup Preferences
- Permission

Script ▾ Help

Availability group name: AlwaysOnAG01

Availability Databases

Database Name
AGDB01
AGDB02

Add Remove

Availability Replicas

Server Instance	Role	Availability Mode	Failover Mode	Connections in Primary Role	Readable Secondary	Session Timeout (seconds)
SQL1	Primary	Synchronous commit	Automatic	Allow all connections	Yes	10
SQL2	Secondary	Synchronous commit	Automatic	Allow all connections	Yes	10
SQL3	Secondary	Asynchronous commit	Manual	Allow all connections	No	10

Add Remove

Connection

Server:
AGVNN

Connection:
SOLIDQ\Administrator

View connection properties

Progress

Ready

OK Cancel

Step-by-step AG

▶ Accessing AG: readable Secondary Replica

simple-queries.sql - SQL2.master (SOLIDQ\Administrator (52)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- AGVNN (SQL Server 11.0.2100 - S
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Prima
        - Availability Replica
        - Availability Databa
        - Availability Group I
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (52))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL2

(1 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)
```

100 %

Query executed successfully. | SQL2 (11.0 RTM) | SOLIDQ\Administrator (52) | master | 00:00:00 | 11 rows

Ready | Ln 5 | Col 3 | Ch 3 | INS

Step-by-step AG

▶ Accessing AG: configure Application Intent

Availability Group Properties - AlwaysOnAG01

Select a page
- General
- Backup Preferences
- Permission

Script ▾ Help

Availability group name: AlwaysOnAG01

Availability Databases

Database Name
AGDB01
AGDB02

Add Remove

Availability Replicas

Server Instance	Role	Availability Mode	Failover Mode	Connections in Primary Role	Readable Secondary	Session Timeout (seconds)
SQL1	Primary	Synchronous commit	Automatic	Allow read/write connections	Read-intent only	10
SQL2	Secondary	Synchronous commit	Automatic	Allow read/write connections	Read-intent only	10
SQL3	Secondary	Asynchronous commit	Manual	Allow all connections	No	10

Add Remove

Connection

Server:
AGVNN

Connection:
SOLIDQ\Administrator

View connection properties

Progress

Ready

OK Cancel

Step-by-step AG

▶ Accessing AG: VNN without Application Intent

Step-by-step AG

- Accessing AG: VNN without Application Intent defaults to read/write

simple-queries.sql - AGVNN.master (SOLIDQ\Administrator (59)) - Microsoft SQL Server Management Studio (Administrator)

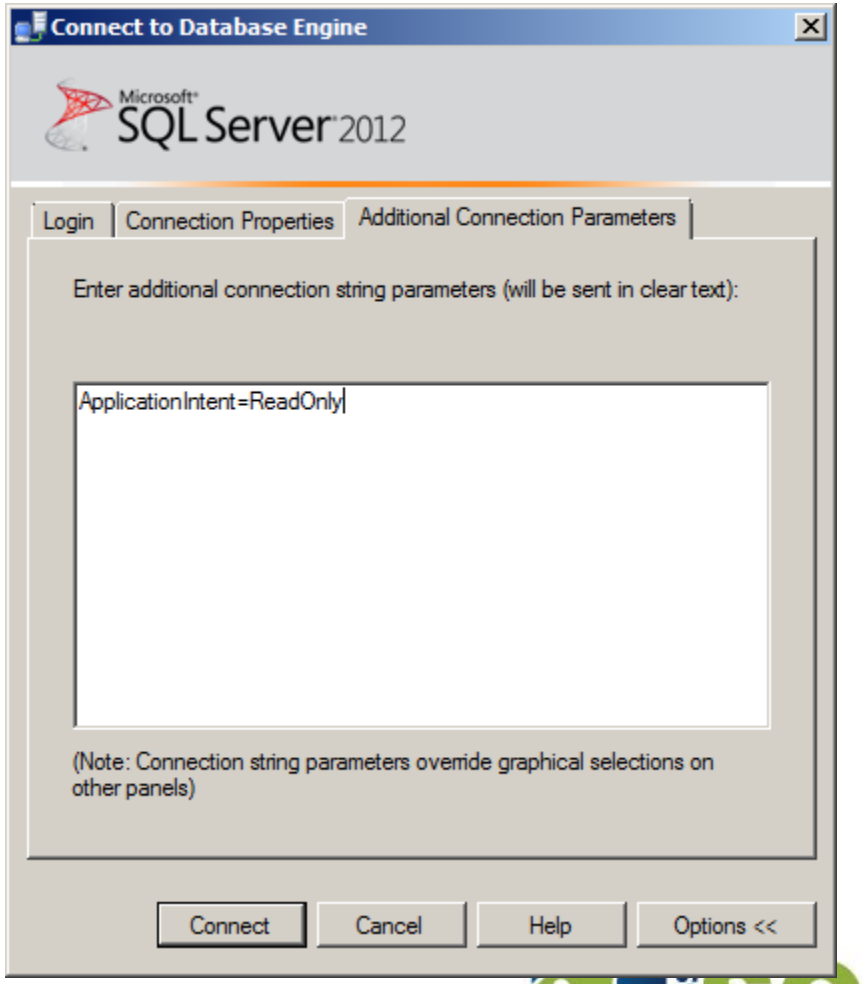
```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

```
Instance
--------------------------------------------------------------------------------
SQL1

(1 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)
```

Query executed successfully. | AGVNN (11.0 RTM) | SOLIDQ\Administrator (59) | master | 00:00:00 | 11 rows

Step-by-step AG

▶ Accessing AG: VNN with Read Only Application Intent



Step-by-step AG

▶ Accessing AG: VNN with Read Only Application Intent still on Primary

simple-queries.sql - AGVNN.master (SOLIDQ\Administrator (61)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- AGVNN (SQL Server 11.0.2100 - S
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Prima
        - Availability Replica
        - Availability Databa
        - Availability Group I
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (61))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL1

(1 row(s) affected)

Msg 979, Level 14, State 1, Line 1
The target database ('AGDB01') is in an availability group and currently does not allow
read only connections.

For more information about application intent, see SQL Server Books Online.
```

100 %

Query completed with errors. | AGVNN (11.0 RTM) | SOLIDQ\Administrator (61) | master | 00:00:00 | 1 rows

Ready | Ln 11 | Col 1 | Ch 1 | INS

Step-by-step AG

▶ Accessing AG: Read Only Routing configuration

configure-replicas.sql - SQL1.master (SOLIDQ\Administrator (62)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

Object Explorer

- SQL1 (SQL Server 11.0.2100 - SOLIDQ\Ac
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01
    - AGDB02
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Primary)
        - Availability Replicas
        - Availability Databases
        - Availability Group Listener
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

configure-replicas.sq...Q\Administrator (62)) | AlwaysOnAG01:SQL2 | AlwaysOnAG01:SQL1 | Dashboard: SQL1

```
-- Configure Read Only Routing URLs
ALTER AVAILABILITY GROUP [AlwaysOnAG01]
MODIFY REPLICA ON N'SQL1' WITH (SECONDARY_ROLE(READ_ONLY_ROUTING_URL=N'TCP://sql1.solidq.loc:1433'))
GO
ALTER AVAILABILITY GROUP [AlwaysOnAG01]
MODIFY REPLICA ON N'SQL2' WITH (SECONDARY_ROLE(READ_ONLY_ROUTING_URL=N'TCP://sql2.solidq.loc:1433'))
GO

-- Configure Read Only Routing lists
ALTER AVAILABILITY GROUP [AlwaysOnAG01]
MODIFY REPLICA ON N'SQL1' WITH (PRIMARY_ROLE(READ_ONLY_ROUTING_LIST=(N'SQL2', N'SQL1')))
GO
ALTER AVAILABILITY GROUP [AlwaysOnAG01]
MODIFY REPLICA ON N'SQL2' WITH (PRIMARY_ROLE(READ_ONLY_ROUTING_LIST=(N'SQL1', N'SQL2')))
GO
```

Messages

```
Command(s) completed successfully.
```

Query executed successfully. | SQL1 (11.0 RTM) | SOLIDQ\Administrator (62) | master | 00:00:01 | 0 rows

Ready | Ln 36 | Col 1 | Ch 1 | INS

Step-by-step AG

▶ Accessing AG: VNN with Read Only Application Intent again on Primary

simple-queries.sql - AGVNN.master (SOLIDQ\Administrator (61)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- AGVNN (SQL Server 11.0.2100 - S
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Prima
        - Availability Replica
        - Availability Databa
        - Availability Group I
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (61))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL1

(1 row(s) affected)

Msg 979, Level 14, State 1, Line 1
The target database ('AGDB01') is in an availability group and currently does not allow
read only connections.

For more information about application intent, see SQL Server Books Online.
```

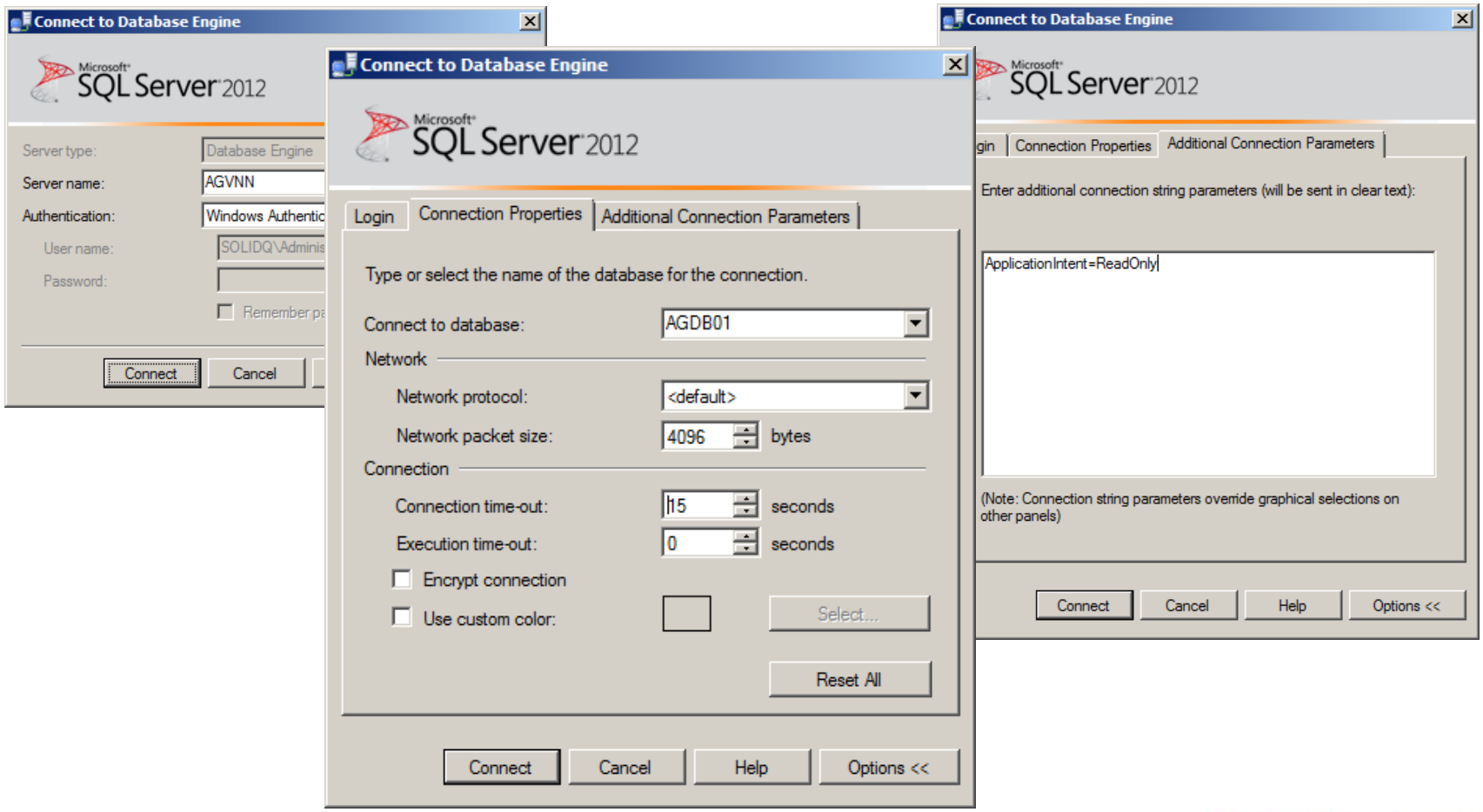
100 %

Query completed with errors. | AGVNN (11.0 RTM) | SOLIDQ\Administrator (61) | master | 00:00:00 | 1 rows

Ready | Ln 11 | Col 1 | Ch 1 | INS

Step-by-step AG

▶ Accessing AG: VNN with Read Only Application Intent and database name



Step-by-step AG

- Accessing AG: Read Only Routing working!

simple-queries.sql - AGVNN.AGDB01 (SOLIDQ\Administrator (52)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | AGDB01 | Execute | Debug

Object Explorer

Connect

- SQL1 (SQL Server 11.0.2100 - SOLIDQ\Ac
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01
    - AGDB02
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Primary)
        - Availability Replicas
        - Availability Databases
        - Availability Group Listener
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (52)) | AlwaysOnAG01:SQL2 | AlwaysOnAG01:SQL1 | Dashboard: SQL1

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL2

(1 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)
```

100 %

Query executed successfully. | AGVNN (11.0 RTM) | SOLIDQ\Administrator (52) | AGDB01 | 00:00:00 | 11 rows

Ready | Ln 4 | Col 29 | Ch 29 | INS

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▶ Read-Only Access Capacity Planning

- Synchronization latency of replica can raise because of the read I/O activity
- Impact on tempdb
  - Temporary statistics
    - Automatically created
    - Gone after failover
  - Read Committed Snapshot Isolation
    - 14 bytes added to rows at primary (on updates)

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▶ Problem with replicas at database level

- Need Manual synchronization for objects outside database scope
  - New logins/users
  - Jobs
- Solution for logins/users
  - Contained databases
  - Problem with **ApplicationIntent=ReadOnly**

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▶ Contained Database

- Allows databases isolation
  - Remove dependecies from instance objects
  - It's a databases level property
  - Only partial support in SQL Server 2012
    - DMV/Extended Event to find non contained entities
- Main usage: database level authentication
  - Login not needed, user is used instead
  - Needs to be permitted at instance level

Advanced AlwaysOn Scenarios

▶ WSFC: Quorum, votes and DR site

- Quorum
  - Majority Node, Majority Node and Fileshare
  - Can’t use Quorum Disk without shared storage
- Windows 2008 e 2008 R2 + QFE
  - Allow to assign voting rights to nodes
  - Avoid scenarios where DR site takes over
  - Common scenario is to always failover manually to DR site i.e. DR site is not considered for HA

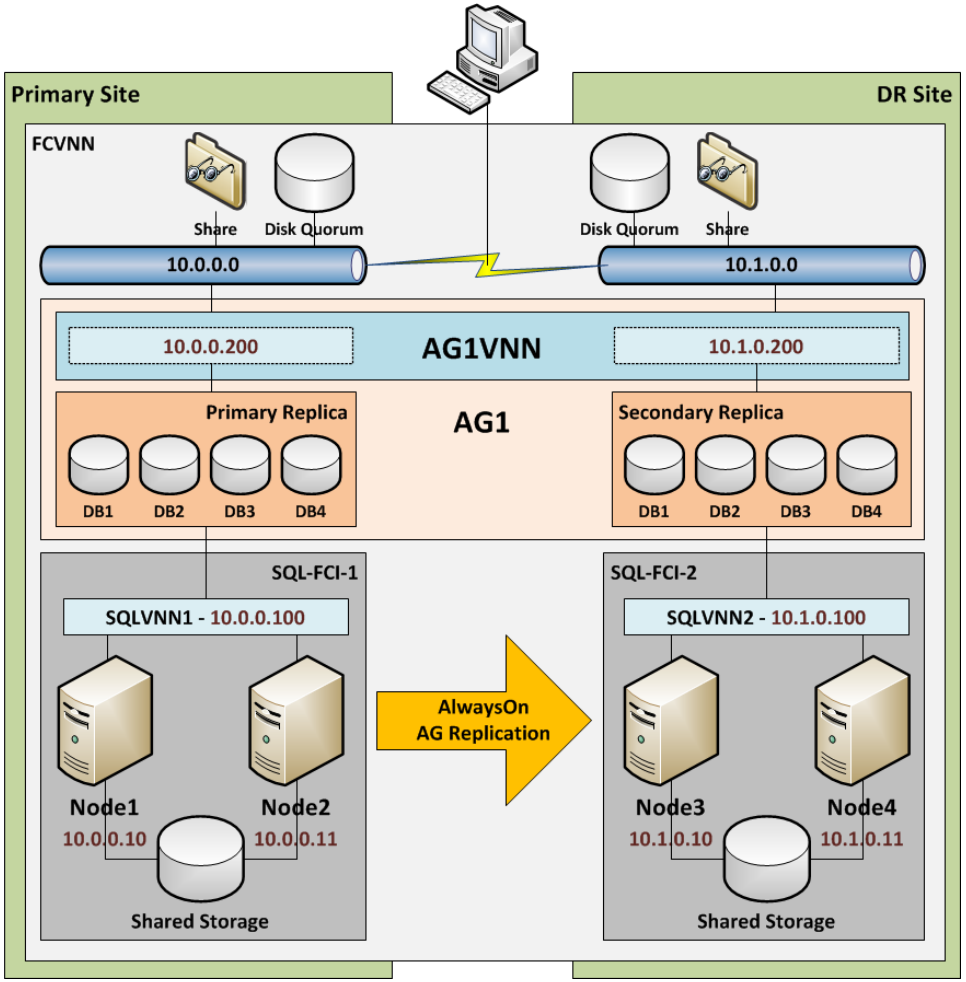
Advanced AlwaysOn Scenarios

▶ WSFC: Asymmetric Storage

- Windows 2008 + QFE or 2008 R2 + SP1
- Disks shared only among some nodes
  - E.g. one SAN on primary site and one SAN on DR site, volumes shared only among local nodes while the cluster stretches across sites
- Allows new Quorum models
  - Asymmetric Disk Majority
  - Node Majority + Asymmetric Disk Majority

Advanced AlwaysOn Scenarios

- Multi-site FCI+AG scenario



Advanced AlwaysOn Scenarios

▶ Topology Example 2

Shared storage, regional and geo secondaries

Asynchronous data movement

HA/DR Editions and Support

▶ Minimum Support for non Enterprise editions

- Server Core
- Log Shipping (except on Express)
- Database Mirroring
  - Deprecated!
  - Synchronous only for BI/Standard
  - Witness only for Web/Express
- Failover Clustering Instance
  - 2 nodes for BI/Standard
- Purely passive nodes licensing as in 2008

SQL Server 2012 Service Pack 1

▶ AlwaysOn related fixes

- Fixes here
  - http://support.microsoft.com/kb/2674319
- CU3 and CU4 fixes not included
  - http://support.microsoft.com/kb/2723749
  - http://support.microsoft.com/kb/2758687
- Problems with Slipstream ISO images
  - http://support.microsoft.com/kb/2783963

SQL Server 2012 Service Pack 1

▶ AlwaysOn new features

- Cross-Cluster Migration of AlwaysOn Availability Groups for OS Upgrade
  - E.g. upgrading to Windows 2012
  - Books Online
    - http://msdn.microsoft.com/en-us/library/cc645581.aspx
  - Whitepaper
    - http://go.microsoft.com/fwlink/?LinkId=265471
    - Still missing as of today!

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▶ AG Main Resources

- Books Online
- SQL AlwaysOn Blog
  - http://blogs.msdn.com/b/sqlalwayson
- CSS SQL Server Engineers
  - http://blogs.msdn.com/b/psssql/archive/tags/alwayson
- SQL CAT
  - http://sqlcat.com

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▶ AG Whitepapers series: AlwaysOn HA/DR architectures and design patterns

- **AlwaysOn Architecture Guide: Building a High Availability and Disaster Recovery Solution by Using AlwaysOn Availability Groups**
  - http://msdn.microsoft.com/en-us/library/jj191711.aspx
- **AlwaysOn Architecture Guide: Building a High Availability and Disaster Recovery Solution by Using Failover Cluster Instances and Availability Groups**
  - http://msdn.microsoft.com/en-us/library/jj215886.aspx
- **AlwaysOn Solution Guide: Offloading Read-Only Workloads to Secondary Replicas**
  - http://msdn.microsoft.com/en-us/library/jj542414.aspx
- **Cross-cluster Migration of AlwaysOn Availability Groups for Operating System Upgrades**
  - http://msdn.microsoft.com/en-us/library/jj873730.aspx
- **Microsoft SQL Server AlwaysOn Solutions Guide for High Availability and Disaster Recovery**
  - http://msdn.microsoft.com/en-us/library/hh781257.aspx
- **Migration Guide: Migrating to AlwaysOn Availability Groups from Prior Deployments Combining Database Mirroring and Log Shipping**
  - http://msdn.microsoft.com/en-us/library/jj635217.aspx
- **Multisite Failover Cluster Instance**
  - http://sqlcat.com/sqlcat/b/whitepapers/archive/2011/12/22/sql-server-2012-alwayson_3a00_-multisite-failover-cluster-instance.aspx
- **SQL Server 2012 AlwaysOn High Availability and Disaster Recovery Design Patterns**
  - http://sqlcat.com/sqlcat/b/msdnmirror/archive/2011/12/22/sql-server-2012-alwayson-high-availability-and-disaster-recovery-design-patterns.aspx

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▶ AG Resources: articles

- **Dashboard**
  - http://www.sqlskills.com/blogs/joe/post/Answering-Questions-with-the-AlwaysOn-Dashboard.aspx
- **Health Model**
  - http://blogs.msdn.com/b/sqlalwayson/archive/2012/02/09/overview-of-the-alwayson-manageability-health-model.aspx
  - http://blogs.msdn.com/b/sqlalwayson/archive/2012/02/13/extending-the-alwayson-health-model.aspx
- **Client Reconnect**
  - http://www.mattmasson.com/index.php/2012/03/alwayson-multi-subnet-failover-and-ssis
- **Failing Over Stack Overflow and Stack Exchange for Sandy**
  - http://blog.serverfault.com/2012/10/30/failing-over-stack-overflow-and-stack-exchange-for-sandy
- **Rick's SQL Server Blog**
  - http://dangerousdba.blogspot.com

Questions & Answers